MODULE 9

Industrial Applications

Steel · refining · ammonia · methanol · fuel cells · transport · storage

Refineries

Hydrotreating removes sulphur from fuels; hydrocracking breaks heavy oil into lighter products.

$$R\text{–}S + 2H_2 \rightarrow R\text{–}H + H_2S \text{ (desulphurisation, simplified)}$$

Largest single user today; replacing grey with green/blue H_2 cuts refinery emissions directly.

Ammonia & fertiliser

Haber–Bosch synthesis of ammonia; ammonia makes urea and other fertilisers.

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

India's green ammonia tenders under the National Green Hydrogen Mission target this use.

Methanol

Syngas or CO_2 + H_2 converted over copper-based catalyst.

$$CO + 2H_2 \rightarrow CH_3OH \cdot CO_2 + 3H_2 \rightarrow CH_3OH + H_2O$$

Chemical feedstock and marine fuel.

Steel (DRI)

Hydrogen reduces iron ore to sponge iron in a shaft furnace, then an electric arc furnace makes steel.

$$Fe_2O_3 + 3H_2 \rightarrow 2Fe + 3H_2O$$

Replaces coke/coal in the blast furnace – water instead of CO_2.

Fuel cells

Electrochemical reverse of electrolysis: H_2 + O_2 → electricity + heat + water.

$$H_2 + \tfrac{1}{2}O_2 \rightarrow H_2O + \text{electricity}$$

PEM for vehicles; SOFC for stationary power.

Heavy-duty transport

Fuel-cell or H_2-combustion trucks and buses; ammonia/methanol for ships.

India's NGHM pilots: 70 hydrogen buses and trucks, 16 refuelling stations (see Hydrogen Vehicle Study deck).

Energy storage

Surplus renewable power → electrolysis → storage → fuel cell or turbine.

Long-duration and seasonal storage; low round-trip efficiency.

Chemical industry

Hydrogenation of oils, production of hydrogen peroxide, pharmaceuticals, electronics.

Small volumes, often high purity.

9.1 India context

India's National Green Hydrogen Mission (launched 4 January 2023, outlay ₹19,744 crore to FY 2029-30) targets about 5 MMT per year of green hydrogen production by 2030, with incentives for

electrolyser manufacturing and green hydrogen production (SIGHT) and pilots in steel, mobility and shipping [R9]. Refining and fertiliser are the obvious first users because they already consume grey hydrogen.

Quiz & revision – Module 9

Answer first, then check the key at the end of the module. Score yourself in the Progress Tracker (Appendix D).

Q1. Reaction for ammonia synthesis:
a) $N_2 + 3H_2 \rightleftharpoons 2NH_3$ **b)** $CO + 2H_2 \rightarrow CH_3OH$ **c)** $C + H_2O \rightarrow CO + H_2$ **d)** $2H_2O \rightarrow 2H_2 + O_2$

Q2. In H2-DRI steelmaking the by-product is:
a) CO_2 **b)** Water **c)** Slag only **d)** Methane

Q3. Largest current hydrogen consumer:
a) Fuel-cell cars **b)** Refining **c)** Aviation **d)** Glass

Q4. Which fuel cell type is common in vehicles?
a) SOFC **b)** PEMFC **c)** MCFC **d)** PAFC

Q5. India's NGHM 2030 production target:
a) 0.5 MMT/yr **b)** ~5 MMT/yr **c)** 50 MMT/yr **d)** 100 MMT/yr

Answer key
Q1 – a: ; **Q2 – b**: ; **Q3 – b**: ; **Q4 – b**: ; **Q5 – b**: